

Does the Benchmark Require the Input? A Falsification Audit for Agent Evaluation

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Abstract

Agent benchmarks are often read causally: a high score is taken as evidence that a system used the current task, plan, graph, or tool state correctly. We test a prior requirement for that interpretation: does the evaluation require the input at all, and is the output paired to the right input? We combine established shortcut diagnostics into a pre-interpretation audit: an out-of-group no-input predictor, a wrong-pairing permutation, and evaluator-provenance checks when references are model-generated. The individual tests are not new; the contribution is an executable admission test for input-conditioned claims and an empirical failure case in which a benchmark deliberately redesigned to escape one shortcut preserves it at a deeper level. On two source-grounded workflow scopes, a scenario-cross-fitted predictor that never receives the case achieves release-set micro-F1 0.919 and 0.589, versus 0.459 and 0.078 for the best measured agents. Re-assigning agent outputs to the wrong cases leaves those scores essentially unchanged. As a public positive control, the same audit on BFCL Multiple behaves oppositely: Gemini 3.8 Flash scores 0.960 with the query, 0.000 when the query is removed, and 0.0049 on average under wrong pairings. A separate evaluator study shows another information-loss mode: Claude Opus 5 agrees with the original evaluator on only 45.5% of latent decision statuses but produces identical scored document sets on all 58 units. These results distinguish benchmark shortcuts, system output collapse, and scoring-projection collapse, and motivate a simple rule: falsify input dependence before attributing an agent score to input-conditioned competence.

1 Introduction

A benchmark score is routinely interpreted as evidence about a system: a model understood a request, an agent used its tools, or an explicit representation improved reasoning on the current instance. That interpretation has a falsifiable prerequisite. The score must depend on the information that the claim says the system used.

This prerequisite is easy to violate. Partial-input baselines expose tasks solvable from annotation artifacts or one side of the input (Kaushik & Lipton, 2018; Poliak et al., 2018; Gururangan et al., 2018); HANS shows that strong NLI systems can succeed for the wrong syntactic reasons (McCoy et al., 2019); simple heuristics expose reading-comprehension questions that do not demand the intended reasoning (Sugawara et al., 2018). More broadly, benchmark design has long been criticized for failing to identify the capabilities attributed to scores (Bowman & Dahl, 2021; Thomas et al., 2026; Joshi et al., 2026). Structured-agent evaluation adds another layer: plans, graphs, tools, model-generated judges, and set-valued outputs can all compress or confound the information being measured.

We encountered this problem while building CasePath, a source-grounded workflow agent. Its first benchmark was static: predict which documents a case requires. We discovered that every measured case in each of two scopes had the same gold checklist. We therefore redesigned the benchmark as paired interventions: add a fact that settles one branch and score which requirements are retracted. The redesign was intended specifically to remove the constant level that broke the static task. It did not. A case-independent release-set predictor

still outperforms every measured agent, and shuffling agent outputs onto the wrong cases barely changes their scores.

The paper asks a narrower and more general question than whether CasePathworks: what must be true before an agent benchmark score can support an input-conditioned capability claim? We use two simple gaps. The input gap compares the evaluated system with a cross-fitted predictor that is denied the evaluation input. The pairing gap compares the observed score with the score after outputs are assigned to the wrong targets within the same inference group. If the first is non-positive, the task can be solved at least as well without the claimed input. If the second is near zero, the evaluated outputs do not demonstrate instance-specific alignment to the targets. Neither test proves construct validity when it passes; either can falsify an input-conditioned interpretation when it fails.

Three findings matter. First, the paired CasePath task fails both tests on two separately authored scopes under a symmetric release-set score. Second, a public tool-use positive control behaves in the opposite direction: on the 200-item BFCL Multiple split (Patil et al., 2025), the same model loses essentially all AST accuracy when its user query is removed, and output permutations destroy its score. Third, evaluator agreement at the scored-output level can conceal large disagreement in latent process state: one independent model family reproduces all final document sets while disagreeing with nearly half of the underlying decision statuses.

We do not claim that input ablations or permutations are novel statistical tools. The contribution is to make them an explicit admission test for agent-evaluation claims, to show why a seemingly stronger paired design can inherit the shortcut it was built to escape, and to release an end-to-end negative record in which apparently significant results are withdrawn after the audit.

2 Related Work

Shortcuts and partial-input baselines. Hypothesis-only, passage-only, and heuristic baselines establish the core diagnostic principle: remove information the task is claimed to require and see what remains (Kaushik & Lipton, 2018; Poliak et al., 2018; Gururangan et al., 2018; McCoy et al., 2019). Sugawara et al. (2018) similarly use simple heuristics to separate easy reading-comprehension questions from ones that require richer reasoning. Shortcut learning generalizes the issue beyond NLP (Geirhos et al., 2020), and recent work finds that scale does not eliminate shortcuts in LLMs (Yuan et al., 2024). We extend the diagnostic to a dynamic set-valued agent task and, importantly, cross-fit the no-input predictor so its policy is learned without the held-out group’s labels.

Benchmark validity. Bowman & Dahl (2021) argue that benchmark performance should imply the in-domain capability the benchmark purports to measure. Thomas et al. (2026) formalize related concerns as evaluation validity, while Joshi et al. (2026) show that representation metrics can produce false conclusions when their structural assumptions fail. Our audit is intentionally operational rather than a new theory of validity: it tries to falsify one concrete interpretation—that a score measures response to the current input—before model differences are interpreted.

Evaluator dependence. Model-generated evaluators can favor related model families; Li et al. (2026) call one such effect preference leakage. We had exactly this confound: the original reference adjudicator belonged to a family also evaluated as an interpreter. We therefore repeat reference construction with independent model families. The result does not validate the labels; instead, it reveals that the final scoring projection can be far more stable than the underlying adjudicated state.

Agent and tool-use benchmarks. BFCL evaluates function/tool calling across simple, multiple, parallel, and agentic settings (Patil et al., 2025). We use its 200-item V4 Multiple split only as a positive control for our audit, not as an official leaderboard submission: we keep BFCL’s published data, possible answers, and AST checker fixed, but run one fixed model

through a JSON function-call prompt so that the only experimental difference is whether the user query is present.

3 An Input-Dependence Audit

Let a benchmark contain grouped instances (x_i, y_i) , an evaluated system output $\hat{y}_i$, and score $S(\hat{y}, y)$. We define two descriptive gaps; classical confidence intervals are not required for the admission decision.

3.1 Input gap

Choose a predeclared class of predictors that cannot access x_i . Fit the strongest member on all groups except one and evaluate it on the held-out group; repeat over groups. Let c_{-g} be the policy learned without group g . The cross-fitted no-input score is

$$S_{\text{noinput}} = S(c_{-g(i)}, y_i), \quad (1)$$

and the input gap is $G_{\text{input}} = S(\hat{y}, y) - S_{\text{noinput}}$. A non-positive gap is sufficient to reject the claim that the score demonstrates useful dependence on the current input.

For CasePath the predicted object is the release set itself and S is micro-F1. On each training fold, the constant set maximizing training micro-F1 is obtained by sorting document frequencies and choosing the prefix length with the highest training micro-F1. For exact-match sensitivity we separately use the modal training release set. These are simple empirical-risk-minimizing baselines, not methodological novelties.

For BFCL the stronger ablation preserves each item’s candidate function schemas but removes only the user query. The model can therefore still exploit tool names, descriptions, parameter schemas, and item-specific candidate sets; it is denied precisely the information whose use the function-calling score is meant to establish.

3.2 Pairing gap

Hold all system outputs fixed and randomly reassign complete outputs across targets, preserving the highest-level grouping used for inference when groups exist. Define

$$G_{\text{pair}} = S(\hat{y}, y) - \mathbb{E}_{\pi} S(\hat{y}_{\pi(i)}, y_i). \quad (2)$$

A small gap does not by itself prove that the benchmark is degenerate; it diagnoses that the evaluated outputs do not demonstrate instance-specific pairing to the targets. In combination with a strong no-input predictor, it separates target regularity from system output collapse.

3.3 Evaluator projection

When a learned evaluator produces a latent state z_i that is deterministically mapped to the scored target $y_i = g(z_i)$, agreement on y need not imply agreement on z . We therefore report evaluator identity and both latent-state and scored-output agreement. This is a provenance/measurement check, not a claim that agreement among model families substitutes for expert ground truth.

4 CasePath: A Paired Redesign That Still Does Not Require the Case

4.1 System and benchmark

CasePathmaps versioned primary-law passages to gated propositions, a process graph, active obligations, required facts, evidence capabilities, and document routes. Branch predicates are interpreted against a case and the final checklist is compiled deterministically from active nodes. A document is released when no active node still requires it. The mechanics

Table 1: All policies predict the same object: the gold release set. No-input policies are selected on other scenarios only. “Best agent” is selected only for display, not used for inference.

Scope	Pairs	Gold sets	No-input μ F1	Best agent μ F1	No-input exact	Best agent exact
Rent increase	28	3	0.919	0.459	0.643	0.071
Termination	49	4	0.589	0.078	0.551	0.531

motivate a natural dynamic claim: an explicit process should support coherent requirement retraction when a new fact settles a branch.

The case corpus is synthetic: 150 generated Swiss-tenancy intake claims, not real claimant data. We evaluate two separately authored scopes. Rent increase contributes 28 complete held-out orig/intervention pairs over six scenarios; termination contributes 49 pairs over eight scenarios. The static reference checklist has exactly one observed value in 30 rent-increase originals and one in 49 termination originals, yielding a case-ignoring static F1 of 1.000 in both scopes. This static failure motivated the paired redesign.

The paired target is the set of documents required before but not after the branch-settling addition. Rent increase has three distinct observed release sets; termination has four. We compare direct checklist prediction (B1), graph-as-context prediction (B3), and the compiled induced graph (B5). All policies are now compared on the same predicted object—the release set—using ordinary micro-F1 and exact match. The earlier volume-corrected metric is retained only as historical context in Appendix A.

4.2 A case-independent predictor is stronger on the same score

Table 1 reports the symmetric comparison. For rent increase, the scenario-cross-fitted no-input release predictor reaches micro-F1 0.919; the strongest agent reaches 0.459. Its exact-match rate is 0.643 versus 0.071 for the best agent. For termination, the micro-F1-optimized no-input predictor reaches 0.589 versus 0.078. Under exact match, a separately cross-fitted modal release set reaches 0.551 versus 0.531 for the best agent. The conclusion therefore does not depend on the historical volume normalization or on comparing policies with different request-set sizes.

One rent scenario contains generated narratives of roughly 167k characters whose intervention is beyond the historical evaluator’s 9,200-character cut. Removing the entire corrupted scenario strengthens rather than creates the result: on the remaining 24 pairs, no-input micro-F1 is 0.972 and B5 is 0.482. We report the unfiltered 28-pair result above to avoid choosing the headline after seeing this sensitivity.

4.3 The agent outputs barely know which case they belong to

Using the same release-set micro-F1, B5 scores 0.459 on rent increase while the mean within-scenario wrong-pairing score is 0.458 (permutation $p = 0.74$). B3 on termination scores 0.078 while its wrong-pairing mean is 0.061 ($p = 0.32$); B5 is identically zero under both. These coarse p -values are not treated as classical inference from six or eight clusters. The descriptive point is that incorrect case-to-output pairings reproduce almost all of the apparent performance.

5 A Public Positive Control: BFCL Requires Its Query

A falsification audit should not reject every benchmark. We therefore freeze an external control on BFCL V4 Multiple (Patil et al., 2025): 200 public items, each presenting multiple candidate functions and exactly one gold call. The exact BFCL data, possible-answer file, and AST checker are pinned to public commit 6ea5797. Gemini 3.8 Flash is run once per item in two matched conditions at temperature zero. The full condition receives the official user query and function schemas. The ablation receives identical schemas but replaces only

Table 2: BFCL positive control. Removing only the user query destroys the same model’s ability to produce the gold tool call.

BFCL condition	AST accuracy	Function-name accuracy	Parse rate
Full query	0.960	0.995	1.000
Query removed	0.000	0.120	0.315

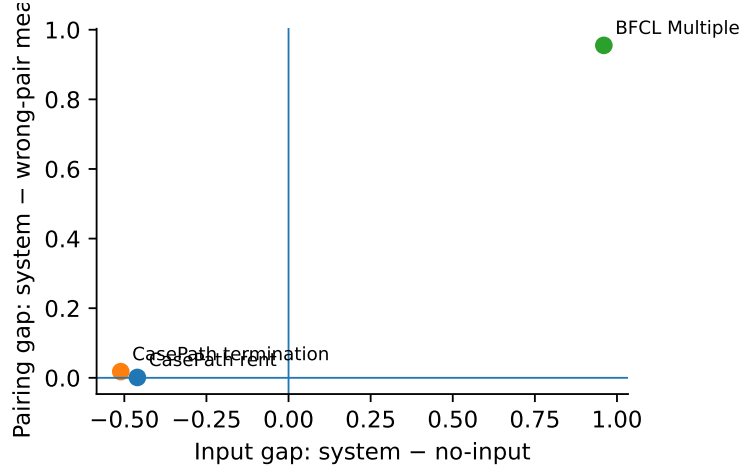


Figure 1: Input and pairing gaps on one common score per evaluation. CasePath’s best measured agents have negative input gaps and near-zero pairing gaps; BFCL Multiple has large positive gaps.

the query with a removal marker. Parse failures count as incorrect and are not semantically retried. This is a prompt-mode positive control, not an official BFCL leaderboard score.

With the query, AST accuracy is 0.960 and function-name selection is 0.995. Removing the query reduces AST accuracy to 0.000 and function-name accuracy to 0.120; only 31.5% of ablated responses even satisfy the fixed call JSON shape. A second falsifier also passes: 5,000 permutations that reassign the full-input model outputs to the wrong BFCL items yield mean AST accuracy 0.0049, 95% range [0, 0.015], and maximum 0.030, versus 0.960 observed. The same audit that rejects the CasePath interpretation therefore gives large positive input and pairing gaps on a public tool-use benchmark.

6 Evaluator Robustness Can Hide State Disagreement

The original rent-increase reference used GPT-5.6 Terra, which also appears among the interpreter models. Before inspecting new judgments we froze a robustness study using Claude Opus 5, Gemini 3.1 Pro, and DeepSeek V4 Pro, three temperature-zero calls per 58 narratives. Missingness rules and the family-balanced consensus rule were committed while calls were running but before judgment content was inspected; we therefore call this a prospectively frozen robustness study with pre-inspection amendments, not an immutable preregistration.

Claude Opus 5 is the load-bearing sensitivity because it has 58/58 unit coverage and zero failed calls. It agrees with the original reference on every final document set and all 28 paired release sets. Yet it agrees on only 45.5% of the 696 individual decision-status comparisons. The deterministic mapping from latent decisions to document unions therefore erases extensive semantic disagreement. High scored-output agreement is not evidence that evaluators agree on the process state the score appears to summarize.

Gemini and DeepSeek are secondary because their semantic-output missingness is substantial. Gemini adjudicates 51/58 units, DeepSeek 50/58, and the prospectively defined three-

family complete-case consensus retains 43 units and 17 complete pairs. Missingness is visibly non-random by scenario; for example the consensus retains zero S4 pairs and one of four S8 pairs. We report these results in Appendix B rather than use them as the primary robustness claim. The full-coverage Opus result is sufficient to show projection coarsening; it does not establish human label validity.

7 What the Audit Admits and What It Does Not

The three checks play different roles. A cross-fitted no-input baseline tests whether target regularity can explain the score without the claimed input. A pairing test asks whether the evaluated system outputs align to the correct instances rather than merely expressing a standing habit. Evaluator provenance asks whether the scored projection conceals instability in the state from which it is derived. We recommend reporting these before attributing a structured-agent score to input-conditioned competence.

The audit is deliberately asymmetric. Passing does not prove construct validity: BFCL’s large gaps show only that this model’s score strongly depends on the user query and correct item pairing, not that BFCL captures every aspect of real tool competence. Failing is more decisive. If a cross-fitted no-input policy beats the evaluated systems on the same predicted object, the score cannot establish that those systems use the input better than the shortcut. If wrong pairings retain the score, the evaluated outputs do not demonstrate case-specific alignment.

None of the individual operations is novel. The methodological contribution is packaging them as a claim-admission procedure for agent evaluation, applying them symmetrically to a failed dynamic benchmark and a public positive control, and preserving the full sequence by which apparently favorable results were invalidated rather than retrospectively cleaning the record.

8 Limitations

The CasePath evidence is narrow: two synthetic scopes in one legal domain generated by one corpus process. It does not establish that legal-document or agent benchmarks generally have this failure. The BFCL experiment is one public benchmark category and one prompt-mode model; it demonstrates specificity of the audit, not broad external prevalence. A larger benchmark survey is the natural next test, but is not required to interpret the present falsification.

The CasePath validity audit was motivated after observing positive-looking held-out results. We therefore make no confirmatory claim about the underlying agent. Scenario counts are small (six and eight), so bootstrap and permutation values are descriptive; the main conclusions use large deterministic score gaps and cross-fitting rather than calibrated asymptotic uncertainty. The rent reference also contains a known long-input truncation defect, which is why we report whole-scenario exclusion sensitivity.

The independent evaluator study uses models, not expert human annotators. Opus’s full-coverage agreement shows that same-family leakage is not needed to reproduce the final document targets, but it does not validate those targets. Finally, historical model runs did not preserve a complete transport-retry ledger, although final per-unit outputs, runner hashes, model identities, prompts, and all current analysis artifacts are committed and hash-bound.

9 Conclusion

We redesigned a benchmark because its static task could be solved without reading the case, then discovered that the paired dynamic replacement preserved the same problem in a less obvious form. On two scopes, cross-fitted no-input release predictors outperform the measured agents on the same micro-F1 and exact-match objects, while wrong pairings preserve almost all agent performance. A public BFCL control behaves in the opposite

direction: removing the query or breaking the pairing destroys the score. The lesson is not that structured agents fail; it is that benchmark scores should not be attributed to input-conditioned reasoning before input dependence is demonstrated. Remove the input, break the pairing, record evaluator provenance, and only then interpret the score.

AI use statement

Generative AI tools were used extensively for research ideation, code generation and review, literature discovery, drafting and editing, construction of source-grounded system artifacts, execution of evaluated model conditions, and model-based reference adjudication. The paper explicitly studies model-generated evaluation as an experimental factor. All headline numerical claims are regenerated by committed deterministic analysis scripts from hash-bound artifacts; AI-assisted prose was checked against those artifacts and cited sources, and unsupported claims were removed or bounded. The authors take responsibility for the final content, including AI-assisted text, code, and artifacts.

Ethics statement

The CasePath claim material is synthetic and contains no real claimant data. The application domain is legal-workflow support, where overstating benchmark validity could create downstream risk if systems were substituted for professional judgment. We therefore do not claim legal correctness or production readiness and preserve negative and failed results. The BFCL data are redistributed under their stated Apache-2.0 license solely as a pinned benchmark slice for reproducibility.

Reproducibility statement

Canonical CasePath analyses are produced from committed JSON artifacts under research/casepath/; every analysis artifact is listed with SHA-256 and byte size in artifacts/MANIFEST.json. The release-set audit, BFCL positive control, evaluator robustness analysis, missingness audit, exact prompts, public BFCL source commit and hashes, and run-identity sidecar are committed. Appendix E gives exact no-network reproduction commands for all analyses after the raw model outputs have been generated.

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A Historical Result and Why It Was Demoted

The original paired analysis used withdrawal recall and then a within-arm random-drop correction for request volume. Under the corrected exact expectation, compiled B5 on rent increase has excess +0.118, direct B1 −0.060, and graph-then-list B3 +0.011. The raw B5–B1 withdrawal-recall difference was +0.553. Four interpreter runs of the same B5 artifact ranged from +0.118 (GPT-5.6 Terra) and +0.108 (Gemini 2.5 Flash) to −0.036 (DeepSeek v3.2) and −0.078 (Claude Haiku 4.5).

We no longer use these cross-policy excess values as the validity headline. The correction is conditional on each policy’s own request set and withdrawal count, so it is useful for diagnosing volume within a policy but is not a clean common-scale comparison across policies with different request sets. The main paper therefore scores every policy directly on the same release-set object. The interpreter table remains useful only as a record that a frozen representation plus different interpreters can induce different standing document habits.

B Evaluator Robustness and Missingness

The robustness run made 522 physical calls (three families × three repeats × 58 units), with reported provider cost \$24.47. Claude Opus 5 had zero failed calls and adjudicated all 58 units. Gemini had 32 failed/unparseable calls and reached the predeclared $\geq 2/3$ unit rule on 51 units; DeepSeek had 37 failures and reached it on 50. The three-family complete-case consensus retained 43 units and 17 pairs.

The missingness audit confirms that the consensus subset is not representative enough to carry the main claim. It retains 0/4 S4 pairs, 4/5 S5, 3/5 S6, 4/5 S7, 1/4 S8, and 5/5 S9. Gemini’s excluded units are much longer on average because three long S8 units fail, while DeepSeek’s exclusions are mostly short S4 units. This is why the main text uses the full-coverage Opus sensitivity and treats the family consensus only as secondary.

On its available units, DeepSeek agrees with the original evaluator on 92% of final document sets and 81.8% of release sets; Gemini agrees on 100% of both on its complete subsets. The three-family consensus agrees on 100% of final document and release sets on its complete cases. These are projection-stability results, not independent proof that the latent adjudications are correct.

C BFCL Positive-Control Protocol

The BFCL V4 Multiple slice, possible-answer file, and AST checker are copied verbatim from public Gorilla commit 6ea57973c7a6097fd7c5915698c54c17c5b1b6c8; their SHA-256 hashes are recorded in external/bfcl/SOURCE.json. The model is google/gemini-3.8-flash, temperature 0, 900 output-token cap, 200 items per condition. Both conditions receive identical function schemas and output instructions; only the official user query is replaced in the ablated condition. The copied BFCL AST checker passes a 200/200 self-test against a schema-aware concrete representative of the official possible-answer encoding.

An initial transport attempt produced 400 OpenRouter HTTP 404 routing rejections because the endpoint selector used google; none reached a provider, used tokens, or incurred cost. Those responses are preserved. The only repair changed the endpoint selector to google-ai-studio, with model, prompts, conditions, scoring and analyses unchanged. The successful 400 semantic calls cost \$0.876567 as reported by the provider. Full-query responses parse 200/200; query-removed responses parse 63/200. Parse failures score incorrect.

D Corrections Preserved in the Scientific Record

The repository preserves several errors because they constrain what can be claimed. A structural-impossibility argument was wrong because it pooled document supply across cases while the compiler recomputes it per case. A cross-scope quote-failure analysis confused unmapped source/quote lists with fabrication and was retracted. A contaminated rent-increase bucket contained termination cases and its saturation statistics were withdrawn. An amendment to the original primary comparison rested on development behavior that no longer reproduces exactly. An early volume baseline used 300 Monte Carlo draws and was replaced with the exact hypergeometric expectation. Finally, an earlier draft attributed 19 document unions to the independent rent reference contract; exhaustive enumeration gives 13, while 19 belonged to a different induced compiler artifact.

E Reproduction Commands

From the repository root, the canonical analyses after raw model execution require no network calls:

```
cd casepath-api
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/verify_artifacts.py
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/release_set_validity_analysis.py
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/evaluation_validity_analysis.py
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/adjudicator_robustness_analysis.py
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/adjudicator_missingness_analysis.py
../runtime/casepath-dev-v2/venv/bin/python \
  ../research/casepath/run_identity_analysis.py
```

The BFCL scorer can be recomputed from its committed raw responses by running bfcl_positive_control.py with the existing raw artifact; figures and tables are regenerated by iclr2027/make_submission_assets.py. All JSON analysis artifacts are hash-bound by artifacts/MANIFEST.json, and verify_artifacts.py fails if the roster or any hash drifts.